Saudi Football

Infrastructure Study

Process documentation & findings

Prepared for SAFF Management

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Companion to: Saudi Football Study presentation deck (30 slides)

Table of contents

1. Executive summary

This study maps Saudi football infrastructure against demand to identify where SAFF investment will most effectively grow participation. It uses GASTAT 2024–25 official data, integrates SAFF accreditation records with a Google Places facilities database covering 1,997 locations across all 13 administrative regions, and is methodologically aligned with UK Sport England and ONS conventions.

Ten research questions were posed at the start of the project. All ten were answered. Six headline findings emerged.

Six headline findings

1. SAFF accreditation: Only 4.6% of clubs and academies are SAFF accredited (22 of 473). Six regions show zero accreditation.

2. Density gap: Pitches per 10,000 people varies 5x. Megaregions (Riyadh, Makkah) sit at 0.23 — one quarter of the UEFA 1.0 benchmark.

3. Inverted correlation: Saudi r = −0.24 between pitch density and PA. UK ONS reports r = +0.49. Saudi inverts the international pattern.

4. Women's PA already at target: National women's PA = 43.1%, above Vision 2030's 40% target. But the gap to men is 23.4 points.

5. Pitch deserts inside megacities: 55% of Riyadh's and 56% of Makkah's urban area sit more than 2 km from any pitch.

6. Sector mix: 52% commercial, 37% public, 11% mixed. Commercial-led, padel-heavy.

What this means for SAFF

- Don't lead with new pitch construction. The data does not support a build-first strategy.
- Lead with women's programming in Eastern, Madinah, and Aseer (the Tier 1 regions from Q10).
- Treat accreditation as a market-shaping lever — at 4.6%, it is not yet a quality signal.
- Use the operating model (open data, female coach pipeline, quarterly KPI review) as the durable infrastructure.

2. Project journey & methodology

Six phases over four weeks

The work was conducted in six phases. This is what was attempted, what worked, and what required workarounds.

Phase	Activity	Output
1	Frame the questions	10 research questions across 4 areas locked in
2	Collect the data	4 data pillars assembled (Facilities, Demographics, PA, SAFF + MOS)
3	Build the database	Postgres on Supabase with views and Arabic fuzzy matching
4	Run the analysis	SQL views + Python scipy for Q3 geospatial, Q9 correlations
5	Validate findings	Calibrated regional youth and women's PA against GASTAT national totals
6	Deliver outputs	Deck, this documentation, Colab notebook

Methodological foundations

Denominators and grain

- Per-10k denominators throughout (Sport England / UEFA convention).
- 13 administrative regions for cross-sectional analyses (Q1, Q2, Q4–Q10).
- 5 cities for geospatial analysis (Q3): Riyadh, Jeddah, Makkah, Madinah, Dammam.
- Youth = under 21 (ages 0–20). Calibrated against GASTAT 2024 national age structure ≈ 9.2M.
- All correlations tested both unweighted and population-weighted; both reported when they diverged.

Women's PA estimation (hybrid method)

GASTAT 2025 publishes national PA by gender (43.06% women, 66.45% men) and 2025 regional PA totals (without gender breakdown by region). To estimate women's PA per region:

> *pa_women_2025_by_region = pa_total_2025_by_region × (pa_women_2021_by_region ÷ pa_total_2021_by_region)*

The result is calibrated so the population-weighted national women's PA equals 43.06% exactly. Confidence: Medium-High.

Q2 stratified sampling

- n = 106 facilities, allocated across 7 strata (35 venues, 32 outdoor pitches, 12 clubs, 12 academies, 5 indoor, 5 sports complexes, 5 stadiums).
- Hand-coded with explicit rubric: school/municipal → public; padel chains, branded academies → commercial; SAFF-affiliated mixed-use → mixed.
- Wilson 95% CI with finite-population correction (FPC = sqrt((N−n)/(N−1))).
- Final: commercial 51.9% (46.1–57.8), public 37.3% (31.7–42.9), mixed 10.7% (7.2–14.3).

Q3 geospatial method

- scipy.spatial.ConvexHull around each city's pitches, with a 1 km outward buffer.
- 500 m grid; Haversine distance from each grid cell to the nearest pitch via cKDTree.
- Three thresholds: 1 km (walk), 2 km (walk/cycle), 5 km (drive).

Q9 correlation tests

Seven hypothesis tests, all persisted in q9_correlations:

Test	X variable	Y variable	Subset	Result
H1	total/10k	PA total %	all 13	r = −0.24 (weighted −0.54)
H2	pitches/10k	PA total %	all 13	r = −0.27 (weighted −0.56)
H3	total/10k	PA women %	all 13	r = −0.44 (weighted −0.70)
H4	total/10k	PA men %	all 13	r = −0.18 (null)
H5	academies/10k under-21	PA total %	all 13	r = +0.19 (null)
H1'	total/10k	PA total %	excl R+M	r = +0.11 (null)
H3'	total/10k	PA women %	excl R+M	r = +0.06 (null)

UK ONS reference: r = +0.49 (positive correlation between facility density and PA). Saudi inverts the pattern.

3. Data sources & quality

Every finding in this study traces back to one of the sources below. Sources are grouped by category, with the data, recency, and how they were used in the deck.

Demand-side data

Source	Data provided	Year	Used for
GASTAT Census 2022 + 2024 estimates	Population by region; youth under 21 per region	2024	Q1 / Q4 / Q7 / Q9 / Q10 denominators
GASTAT Physical Activity Statistics 2025	National + regional PA totals; gender breakdown national	2025	Q8 / Q9 / Q10
GASTAT Sports Practice Survey 2021 (sheet 2-2)	Within-region gender PA ratios	2021	Q8 women's PA per region (hybrid estimate)

Supply-side data

Source	Data provided	Year	Used for
Google Places API (custom collector)	1,997 facilities with lat/lng, type, rating	Apr 2026	Q1 / Q3 / Q4 / Q5 / Q6 / map
SAFF Accredited Clubs & Academies PDF (OCR'd)	69 entries (clubs + academies)	Oct 2023	Q6 accreditation rate
MOS Open Data — commercial halls register	3,116 halls, 1,175 women-flagged	2024	Q7 women's facilities by region

Saudi women's football (specialized sources)

Source	Data point	Year
SAFF VP Lamia Bahaian, FIFA Women's Football Convention	374 → 694 registered women's growth; 1,000+ female coaches; 90 referees	2023
NEOM / SAFF Women's Football Report	+195% pro player growth since 2018; 77,000 girls in Schools League	Jan 2025
SAFF Saudi Women's Premier League	10 clubs in 2024-25 SWPL	2024-25

Methodology references

Reference	Used for
UK ONS Sport Participation methodology	Q9 cross-sectional regression replication
UK Sport England / UEFA per-10k denominators	Q1 / Q4 grain decision
Wilson 95% CI with finite-population correction	Q2 stratified sampling
scipy.spatial ConvexHull + cKDTree (Haversine)	Q3 pitch desert geospatial computation

pg_trgm + custom arabic_normalize() in PostgreSQL	SAFF-to-facilities fuzzy matching

Benchmark country sources (federation primary data)

Country	Source	Data point used
Germany	DFB Mitgliederstatistik 2024/25 (published July 2025)	23,868 clubs · 2.38M active players · 99,879 adult women
Japan	JFA registered players (Statista citation, Mar 2024)	834,420 registered players
Türkiye	TFF licensed players (latest TFF figure)	466,445 licensed players · 24 licensed clubs 2024-25
Saudi Arabia	SAFF / Saudipedia 2024	170 clubs across 4 divisions · 18 SPL

Note on benchmark gaps: Q2 (public/commercial mix), Q3 (pitch deserts), Q5 (rating reliability), and Q7 (women's facilities) had no comparable data available in international sources for Germany, Japan, or Türkiye in 2024-25 reporting. These are reported as Saudi-specific findings.

Data quality scorecard

Source	Records	Recency	Coverage	Match	Confidence
Facilities (Google Places)	1,997	Apr 2026	96.3%	—	High
Demographics (GASTAT)	13	2024	100%	—	High
PA stats (GASTAT)	13	2025	100%	—	Med-High
SAFF accreditation	69	Oct 2023	100%	32%	Medium
MOS commercial halls	3,116	2024	100%	—	High

Match column reflects how many SAFF entries linked to the facilities table. The 32% rate means 22 of 69 SAFF entries matched; the other 47 are mostly small commercial academies and women's clubs not visible in Google Maps.

4. Findings — Q1 to Q10

Q1 · Pitches per 10,000 people

National average: **0.43 pitches per 10k**

Range: **1.18 (Al-Baha, Northern Borders) → 0.23 (Riyadh, Makkah)**

Megaregions sit at one quarter of the UEFA 1.0 "well-served" benchmark. Only Al-Baha and Northern Borders clear that bar.

Source: Google Places API (Apr 2026) ÷ GASTAT 2024 population estimates. UEFA 1.0 benchmark from UK Sport England.

Q2 · Public, commercial, mixed split

Commercial: **52% (46.1–57.8) — padel chains, branded academies, English-named businesses**

Public: **37% (31.7–42.9) — schools, municipal courts, government complexes**

Mixed: **11% (7.2–14.3) — SAFF-affiliated clubs running both members' programs and paid academies**

Per-stratum: academies ~100% commercial, outdoor pitches ~90% public, football venues ~88% commercial (the padel boom).

Source: Hand-coded stratified sample of n=106 facilities, May 2026, with Wilson 95% CI.

Q3 · Pitch deserts in 5 major cities

City	Pitches	>1 km desert	>2 km desert	>5 km desert
Riyadh	86	83.8%	55.4%	9.7%
Makkah	62	81.8%	56.4%	18.3%
Jeddah	71	78.6%	50.0%	13.2%
Dammam	66	65.2%	28.9%	0.1%
Madinah	58	65.3%	27.7%	1.8%

Two cities stand out: Makkah (worst at the 5 km threshold — 18.3% of urban area is more than a 10-minute drive from any pitch) and Riyadh (worst at the 1 km threshold — 83.8%). Dammam and Madinah are best-served per their footprint.

Source: Geospatial computation on facilities lat/lng. ConvexHull + 1 km buffer + 500 m grid + Haversine distance via cKDTree. Euclidean distance overestimates accessibility (real walking is 1.2–1.4× longer on networks).

Q4 · Academies per 10,000 youth (under 21)

National average: **0.28 academies per 10k youth (under 21)**

Top regions: **Najran 0.52, Qassim 0.46, Hail 0.36**

Bottom regions: **Riyadh 0.15, Al-Jouf 0.15, Al-Baha 0.00**

Megaregions remain under-supplied — same pattern as Q1.

Source: Google Places API academies ÷ GASTAT youth under 21 (calibrated to national 9.2M from 2022 census age-band structure).

Q5 · Google rating reliability

Coverage: **96.3% of facilities have a Google rating (1,924 of 1,997)**

Reliable subset: **47% have 50+ reviews — these average 4.21★**

Below 10 reviews: **Ratings are statistical noise — should be ignored**

By type: Football academy 4.44★ (highest), Football venue 4.28, Sports complex 4.21, Football club 4.18, Indoor pitch 4.17, Stadium 4.16, Outdoor pitch 4.09 (lowest, mostly public neighborhood courts).

Source: Google Places API rating + reviews_count fields, April 2026.

Q6 · SAFF accreditation rate

National rate: **4.6% — only 22 of 473 clubs and academies**

Zero-accreditation regions: **6 of 13 (Hail, Northern Borders, Aseer, Al-Baha, Al-Jouf, Madinah)**

Top region: **Najran at 11.1%**

Accreditation is not yet a usable consumer-side market signal at scale.

Source: SAFF Accredited Clubs & Academies PDF, October 2023 (OCR'd from a ZIP of three scanned JPEGs). Match rate to facilities table: 32%.

Q7 · Women's facilities by region

Total: **1,175 women halls nationally**

Highest density: **Riyadh 5.33 per 100k**

Lowest density: **Northern Borders 0.98 per 100k**

5x spread — same magnitude as total facility density.

Source: MOS Open Data, commercial halls register (2024). 3,116 total halls, 1,175 women-flagged.

Q8 · Women's PA versus infrastructure

Counter-intuitive result: megaregions have the HIGHEST women's PA but the LOWEST hall density per capita.

Riyadh women's PA: **45.8% — second highest**

Makkah women's PA: **48.7% — highest**

National women's PA: **43.1% — already above Vision 2030 40% target**

Football infrastructure is not what is driving women's PA in big cities — gym chains, walking, and transit-related activity dominate.

Source: GASTAT 2025 PA national + 2021 Sports Practice Survey within-region gender ratios (hybrid estimate, calibrated to 43.06% national).

Q9 · PA × pitch density correlation

Saudi r (unweighted): **−0.24**

Saudi r (population-weighted): **−0.54**

Saudi r for women specifically (weighted): **−0.70**

UK ONS reference r: **+0.49**

Saudi inverts the international pattern. Excluding the megaregions reverses the sign but stays statistically null. Implication: building more pitches in regions that already have plenty (Al-Baha, Najran, Qassim) will not lift PA.

Source: 7 hypothesis tests on q9_correlations table; UK reference: ONS Sport Participation methodology 2018-22.

Q10 · Investment recommendation

Tier classification combining women's PA gap × inactive women count × women halls density.

Tier	Regions	Logic
Tier 1 (primary, 3 regions)	Eastern (5.6M, 37.3% women PA, ~960k inactive), Madinah (2.3M, 36%), Aseer (2.2M, 38%)	Low women's PA + large absolute populations of inactive women = highest opportunity
Tier 2 (scale, 6 regions)	Riyadh, Makkah, Jizan, Qassim, Tabuk, Hail	High women's PA already but huge inactive populations make modest gains nationally significant
Tier 3 (defer, 4 regions)	Al-Jouf, Najran, Al-Baha, Northern Borders	Populations too small (<700k each) for meaningful national impact

Operating frame: "convert under-utilized capacity for women," not "build more pitches."

Source: Q10 priorities view combining Q1, Q4, Q7, Q8 outputs. Tier thresholds: women's PA below 40% AND inactive women count above 300k.

5. Recommendations

All seven recommendations are grounded in the Q1–Q10 statistics, not in generic best-practice.

1. Re-tag, don't build

0.43 pitches/10k nationally (Q1) and r = −0.24 PA correlation (Q9) show new construction will not lift PA. Convert under-utilized capacity in well-served regions for women's hours.

2. Lead with women's programming in 3 regions

Eastern (5.6M, 37.3% women PA), Madinah (2.3M, 36%), and Aseer (2.2M, 38%) hold the largest absolute opportunity (Q10). Recruit 50 female coaches there in 24 months.

3. Treat SAFF accreditation as a market-shaping lever

4.6% accreditation rate (Q6) is not a quality signal. Set a 5-year target of 25%, starting with the 6 zero-accreditation regions.

4. Fill desert pockets in Riyadh and Makkah

55% / 56% of urban area beyond 2 km from any pitch (Q3). Re-tag existing pitches and add school partnerships before greenfield construction.

5. Build the female coach pipeline

1,000+ qualified female coaches and 90 referees nationally (SAFF, 2023). Concentrate them in Tier 1 regions. Partner with the Saudi Women's Premier League to seed academies in Madinah, Eastern, Aseer.

6. Use Google ratings as soft quality signal — 50+ reviews only

96.3% of facilities are rated; 47% have 50+ reviews at a mean of 4.21★ (Q5). Build SAFF-verified badges to displace noisy ratings below 10 reviews.

7. Don't replicate Riyadh's 5.33 women halls per 100k elsewhere

Prioritize Northern Borders (0.98) and Aseer (1.26) for women's-specific facilities (Q7). Capacity is uneven across regions, not absent.

6. Benchmark — Saudi vs Germany / Japan / Türkiye

All benchmark figures use 2024–25 federation primary data.

Football fields

Metric	Saudi	Türkiye	Japan	Germany
Total registered clubs	170	24	1,200	23,868
Top-tier pro clubs	18	18	20	18
Population (millions)	35.3	85.5	124	83.4

Saudi: SAFF / Saudipedia 2024 (170 clubs across all 4 divisions). Germany: DFB Mitgliederstatistik 2024/25. Türkiye: TFF licensed clubs 2024-25. Japan: JFA / J-League 2024.

Academies & pipeline

Metric	Saudi	Türkiye	Japan	Germany
Registered players	30,000	466,445	834,420	2,380,000
Per 1,000 population	0.85	5.50	6.73	28.50

Saudi at 0.85 per 1,000 versus Germany at 28.50 — a 33x gap. Türkiye 5.50 and Japan 6.73 are the realistic peer benchmarks.

Germany: DFB Mitgliederstatistik 2024/25. Japan: JFA via Statista, March 2024. Türkiye: TFF latest published figure. Saudi: SAFF data via project files.

Women's football

Metric	Saudi	Germany
Adult registered women players	694 (2022)	99,879 (2024/25)
Growth in registered women	374 → 694 (+106% YoY)	+10% girls' teams 2024/25
Schools-level participation	77,000 girls (2024)	119,000 girls ≤16 (2024/25)
Women's top-tier league clubs	10 (SWPL 2024-25)	12 (Frauen-Bundesliga)
Pro player growth since 2018	+195%	—

Saudi: SAFF VP Lamia Bahaian (FIFA Women's Football Convention 2023), NEOM/SAFF Women's Football Report (Jan 2025). Germany: DFB Mitgliederstatistik 2024/25 (published July 2025). Japan and Türkiye are excluded from this comparison because their federations have not published comparable adult women player counts in 2024-25 reporting.

7. Limitations & caveats

Coverage limitations

- Google Places API likely undercounts informal and school pitches, especially in lower-income areas. Total facility counts are conservative.
- Phone number and website fields are 100% empty in the facilities database (the Places collector was run without --details).
- The SAFF October 2023 list is the most recent published; SAFF may have expanded accreditation since.
- 32% SAFF-to-facilities match rate. The unmatched 47 entries are mostly small commercial academies and women's clubs not visible in Google Maps — a real coverage gap, not a fuzzy-matcher failure.
- Türkiye benchmark uses TFF's latest published figure (466,445 licensed players); TFF does not regularly update detailed facility statistics.

Estimation limitations

- Women's PA per region is a hybrid estimate (2025 regional total × 2021 within-region gender ratio). The 2024–25 gender gap may have narrowed unevenly.
- Youth under 21 by region calibrated from 2022 census age structure scaled to 2024 GASTAT national totals. Sub-region fertility differences are smoothed out.
- Q2 stratified sample carries ±5pp 95% CI on the overall split; per-stratum CIs are wider (±9pp).

Methodological caveats

- Q3 uses Euclidean distance, not network distance. Real walking distance is 1.2–1.4× longer, so the 2 km threshold is closer to 1.5 km of network distance.
- Q9 has only n=13 regions — much coarser than ONS UK's hundreds of Local Authorities. Confidence intervals on the correlation are wide.
- The Saudi-specific findings (Q2, Q3, Q5, Q7) cannot be benchmarked because comparable data is not published in any of Germany, Japan, or Türkiye.

8. Slide-by-slide guide

This is the section to consult during the presentation if a question comes up about a specific slide.

Slide 1

Cover

Summary

Title "Saudi Football Study" with subtitle "Mapping supply against demand to grow Saudi football, prepared for SAFF management." Prepared by Abdulrahman Bajunaid.

Source(s)

Cover slide — no data.

Slide 2

Table of contents

Summary

- 01 Project journey
- 02 Methodology
- 03 Data quality score
- 04 Key insights
- 05 Recommendations
- 06 Benchmark

Source(s)

Navigation slide — no data.

Slide 3

01 — Project journey (section divider)

Summary

Marker for the project-journey section.

Source(s)

Section header — no data.

Slide 4

Project journey: 6 phases

Summary

Six-phase narrative: frame the questions, collect the data, build the database, run the analysis, validate findings, deliver outputs.

Source(s)

Author-defined process narrative for the four-week project.

Slide 5

02 — Methodology (section divider)

Summary

Subtitle: Data sources, denominators, decisions.

Source(s)

Section header — no data.

Slide 6

How we approached it

Summary

- Supply vs Demand — maps nearly 2,000 facilities + SAFF accreditations against GASTAT 2024-25 population and activity data.
- Benchmarking — uses per-10k denominators (Sport England / UK ONS).
- Granularity — 13 administrative regions + 5 cities for geospatial.
- Correlation — both unweighted and population-weighted.
- Key result — failed to replicate UK ONS r = +0.49.

Source(s)

UK Sport England + ONS conventions for the denominator. The map illustration is stylized; analysis covers all 13 regions.

Slide 7

Three data pillars

Summary

- Facilities database — 1,997 records via Google Places API.
- Demographics & PA — GASTAT 2024 population, GASTAT 2025 PA stats, GASTAT 2021 sports practice survey.
- SAFF + MOS — 69 SAFF entries, 3,116 MOS halls (1,175 women).

Source(s)

Each pillar is one upstream dataset, combined into a Supabase Postgres database.

Slide 8

03 — Data quality score (section divider)

Summary

Subtitle: Confidence per source, gaps, caveats.

Source(s)

Section header — no data.

Slide 9

Data quality score per source

Summary

Five-row table covering Facilities, Demographics, PA stats, SAFF accreditation, MOS commercial halls — with records, recency, coverage %, match rate, confidence rating, and notes.

Source(s)

- Facilities — Google Places API custom collector, April 2026.
- Demographics — GASTAT Census 2022 + 2024 estimates.
- PA stats — GASTAT 2025 + 2021 hybrid estimate.
- SAFF — SAFF Accredited Clubs PDF, October 2023.
- MOS — MOS Open Data commercial halls, 2024.

Slide 10

04 — Key results (section divider)

Summary

Subtitle: Diagrams and insights for Q1 to Q10.

Source(s)

Section header — no data.

Slide 11

Where the supply sits

Summary

- 1,997 facilities mapped across 13 regions (green).
- 4.6% SAFF accredited (red — only 22 of 473).
- Map: static snapshot of the 1,997 facility points on a Saudi outline.

Source(s)

1,997 = Google Places API count (April 2026). 4.6% = 22 SAFF-matched ÷ 473 total clubs+academies in the facilities table.

Slide 12

Q1 — Pitches per 10,000 people

Summary

National average 0.43 (red). Range 1.18 down to 0.23 (red). Megaregions at one quarter of the UEFA 1.0 benchmark. Recommendation: prioritize new pitch construction in Riyadh and Makkah; avoid further expansion in Al-Baha and Northern Borders.

Source(s)

Numerator: Google Places API (1,997 facilities, April 2026). Denominator: GASTAT 2024 population estimates. Benchmark: UEFA 1.0 from UK Sport England.

Slide 13

Q2 — Public, commercial, mixed split

Summary

Commercial 52% (red), public 37% (red), mixed 11% (red). All carry ±5pp 95% CI. Recommendation: leverage the dominant commercial sector (52%) — padel chains and branded academies — through re-tagging schemes.

Source(s)

Hand-coded stratified sample of n=106 facilities. Wilson 95% CI with finite-population correction. Coding rubric: school/municipal → public; padel/branded → commercial; SAFF-affiliated → mixed.

Slide 14

Q3 — Pitch deserts in 5 major cities

Summary

At 2 km threshold: Riyadh 55% (red), Makkah 56% (red), Jeddah 50% (red); Dammam 29% (green), Madinah 28% (green). Recommendation: prioritize Riyadh and Makkah for desert-filling.

Source(s)

scipy.spatial.ConvexHull around city pitches + 1 km buffer + 500 m grid + Haversine distance via cKDTree. Three thresholds: 1 km (walk), 2 km (walk/cycle), 5 km (drive).

Slide 15

Q4 — Academies per 10,000 youth (under 21)

Summary

National avg 0.28 (green). Najran 0.52 (green) and Qassim 0.46 (green) lead. Riyadh 0.15 (red), Al-Jouf 0.15 (red), Al-Baha 0.00 (red) trail. Recommendation: seed academies in Al-Baha and Riyadh/Al-Jouf before expanding in Najran and Qassim.

Source(s)

Numerator: Google Places API academies, April 2026. Denominator: GASTAT youth under 21, calibrated to national 9.2M (sum of age bands 0-4, 5-9, 10-14, 15-19, plus 1/5 of 20-24).

Slide 16

Q5 — Google rating reliability

Summary

96% of facilities are rated (green). Reliable subset (50+ reviews) covers 47% (green) — these average 4.2★ (green). Below 50 reviews, ratings are noise. Recommendation: use 50+ reviews as the operational threshold; build a SAFF-verified badge.

Source(s)

Google Places API rating + reviews_count fields, April 2026.

Slide 17

Q6 — SAFF accreditation rate

Summary

4.6% (red) national rate — only 22 of 473 (red). 6 of 13 regions at zero (red). Najran leads at 11.1% (green). Recommendation: treat accreditation as a market-shaping lever; 5-year target of 25% starting with the 6 zero regions.

Source(s)

SAFF Accredited Clubs & Academies PDF, October 2023 (OCR'd from a ZIP of three scanned JPEGs). 32% match rate to facilities table via PostgreSQL pg_trgm with custom arabic_normalize() function.

Slide 18

Q7 — Women's facilities by region

Summary

1,175 (green) women halls nationally. Riyadh leads at 5 per 100k (green). Northern Borders lowest at 0.98 (red). 5x spread mirrors total facility density. Recommendation: prioritize Northern Borders and Aseer for women's-specific facilities.

Source(s)

MOS Open Data commercial halls register (2024). 3,116 total halls of which 1,175 are women-flagged.

Slide 19

Q8 — Women's PA versus infrastructure

Summary

Megaregions show highest women PA (45-49%, green) but lowest hall density per capita. National women PA 43% (green) — already above Vision 2030 40% target. Recommendation: re-tag programming, allocate women-only hours, recruit female coaches.

Source(s)

GASTAT 2025 national women's PA = 43.06%. Per-region women's PA = hybrid 2025 × 2021 estimate, calibrated to national. Halls per region: MOS 2024.

Slide 20

Q9 — PA × pitch density correlation

Summary

Saudi r = −0.24 (pop-weighted −0.54, red). Inverts UK ONS r = +0.49 (green). For women specifically r = −0.70 weighted (red). Recommendation: replicate the megaregion mix — gyms, walkable streets, mixed-use facilities — in Eastern, Madinah, and Aseer instead of defaulting to pitch builds. Riyadh and Makkah hit the highest women's PA on the lowest pitch density.

Source(s)

7 hypothesis tests on n=13 regions, persisted in q9_correlations table. UK reference: ONS Sport Participation methodology 2018-22.

Slide 21

Q10 — Where should SAFF invest first?

Summary

Tier 1: Eastern (5.6M, 37.3% women PA red, ~960k inactive), Madinah (2.3M, 36% red), Aseer (2.2M, 38% red). Recommendation: Tier 1 first; convert capacity, seed SWPL-affiliated academies, recruit 50 female coaches in 24 months.

Source(s)

Q10 priorities view combines pop-weighted women's PA gap × inactive women count × women halls density. Inactive women count = (1 − pa_women) × women 18+ population (GASTAT 2024).

Slide 22

05 — Recommendations (section divider)

Summary

Subtitle: Seven actions grounded in Q1–Q10.

Source(s)

Section header — no data.

Slide 23

Recommendations (single slide, 7 actions)

Summary

- 1. Re-tag, don't build (Q1, Q9)
- 2. Lead with women's programming in 3 regions (Q10)
- 3. Treat SAFF accreditation as a market-shaping lever (Q6)
- 4. Fill desert pockets in Riyadh and Makkah (Q3)
- 5. Build the female coach pipeline
- 6. Use Google ratings as soft quality signal — 50+ reviews only (Q5)
- 7. Don't replicate Riyadh's 5 women halls per 100k elsewhere (Q7)

Source(s)

All seven derived from Q1–Q10 findings on prior slides. Female coaches and referees figure (1,000+ and 90) is from SAFF VP Lamia Bahaian's FIFA Women's Football Convention 2023 statement.

Slide 24

06 — Benchmark (section divider)

Summary

Subtitle: Saudi vs Germany / Japan / Türkiye in 3 categories.

Source(s)

Section header — no data.

Slide 25

Benchmark — Football fields

Summary

Total registered clubs: Saudi 170, Türkiye 24, Japan 1,200, Germany 23,868. Top-tier pro clubs all 18-20.

Source(s)

Saudi: SAFF / Saudipedia 2024. Germany: DFB Mitgliederstatistik 2024/25 (23,868 clubs). Japan: JFA / J-League 2024 (~1,200 clubs). Türkiye: TFF licensed clubs 2024-25 (24).

Slide 26

Benchmark — Academies & pipeline

Summary

Registered players per 1,000 people: Saudi 0.85, Türkiye 5.50, Japan 6.73, Germany 28.50. Saudi at 33x gap to Germany; Türkiye and Japan are nearer peers.

Source(s)

- Germany: DFB 2024/25 (2.38M ÷ 83.4M).
- Japan: JFA via Statista, March 2024 (834,420 ÷ 124M).
- Türkiye: TFF latest figure (466,445 ÷ 85.5M).
- Saudi: SAFF data via project files.

Slide 27

Benchmark — Women's football

Summary

Saudi adult registered women: 694 (2022). Schools league: 77,000 girls (2024, +106% YoY in registered). +195% pro player growth since 2018. Germany: 99,879 adult women (2024/25).

Source(s)

- Saudi 374 → 694: SAFF VP Lamia Bahaian, FIFA Women's Football Convention 2023.
- Saudi 77,000 schools + 195% growth: NEOM/SAFF Women's Football Report (Jan 2025).
- Germany 99,879: DFB Mitgliederstatistik 2024/25.
- Japan and Türkiye excluded — federations have not published comparable adult women player counts.

Slide 28

Vision 2030 quote

Summary

Closing aspirational quote framing the three pillars (pipeline, pitches, women's game) aligned with Vision 2030 Quality of Life Programme.

Source(s)

Aspirational closing slide — no data.

Slide 29

Thank you

Summary

Title "Thank you" with author credit. Footer: "Questions and discussion."

Source(s)

Closing slide — no data.

Slide 30

References & sources

Summary

Consolidated index of every source used in the deck, organized by Demand-side, Supply-side, Methodology, Benchmark country sources, Saudi women's football, and Database.

Source(s)

Reference slide — see Section 3 of this document for the full source detail.

9. Deliverables & database connection

Deliverables

- This documentation (.docx)
- Saudi Football Study presentation deck (.pptx, 30 slides)
- Saudi_Football_Analysis.ipynb — Colab notebook with all analytical sections, charts, and CSV exports
- Supabase project Football_Fields_Mapping_SA — all data, views, and queries

Database connection

Setting	Value
Project name	Football_Fields_Mapping_SA
Project ID	hdcqusmfyovcvtdlawbm
Region	ap-southeast-1
Postgres version	15
REST base URL	https://hdcqusmfyovcvtdlawbm.supabase.co/rest/v1/
Publishable key (read-only, safe to embed)	sb_publishable_uhXX_xRAVZ2s6nKtJiWCmw_8UPPQQTu

Data dictionary — primary tables and views

Object	Rows	Purpose
facilities	1,997	All Google-Places-derived football facilities
regional_demographics	13	Population, youth under 21 per region
regional_pa_2025	13	PA total + women + men estimates per region
saff_accredited	69	SAFF Oct 2023 list, mapped to facilities (22 matched)
regional_women_facilities	13	MOS commercial halls women / total / per 100k
q2_sample_coding	106	Hand-coded sample with rationale per row
q3_pitch_deserts	15	5 cities × 3 thresholds, % desert + area km²
q9_correlations	7	Pearson + Spearman + weighted r per hypothesis
v_regional_master (view)	13	Pre-joined master with all 30+ region metrics
v_q10_priorities (view)	13	Tier classification per region (Tier 1/2/3)
v_data_quality (view)	8	Data quality scorecard

Reproducibility

All analyses are reproducible from the persisted tables. Q9 correlations re-run from regional_demographics + regional_pa_2025 + facilities counts. Q3 desert analysis re-runs from facilities lat/lng for the 5 city_id values. Q2 mix re-runs from q2_sample_coding via the q2_population_estimates view.